a) Esercizio 1

$v = 2\vec{i} - 3\vec{j} + \vec{k}$ $\quad w = \frac{5}{3}\vec{i} - \frac{5}{2}\vec{j} + \frac{5}{6}\vec{k}$

v e w sono ortogonali se $\langle v, w \rangle = 0$

$$\langle v, w \rangle = 2 \cdot \frac{5}{3} - 3 \cdot \left(-\frac{5}{2}\right) + 1 \cdot \frac{5}{6} =$$

$$= \frac{10}{3} + \frac{15}{2} + \frac{5}{6} = \frac{20 + 45 + 5}{6} = \frac{70}{6} = \frac{35}{3} \neq 0$$

v e w non sono ortogonali

v e w sono paralleli se $v \times w = 0$

$$v \times w = \begin{vmatrix} i & j & k \\ 2 & -3 & 1 \\ \frac{5}{3} & -\frac{5}{2} & \frac{5}{6} \end{vmatrix} = \vec{i}\left(-\frac{15}{6} + \frac{5}{2}\right) - \vec{j}\left(\cancel{2}\frac{5}{\cancel{6}_3} - \frac{5}{3}\right) +$$

$$+ \vec{k}\left(\cancel{2}\left(-\frac{5}{\cancel{2}}\right) + \cancel{3} \cdot \frac{5}{\cancel{3}}\right) =$$

$$= \vec{i}\,0 - \vec{j}\,0 + \vec{k}\,0$$

$v \parallel w$

b) Sia $u = -\vec{i} + 2\vec{j} + 3\vec{k}$.
Per determinare se è complanare ad v e w, occorre verificare se il prodotto misto è nullo

$$\langle u, v \times w \rangle = \begin{vmatrix} -1 & 2 & 3 \\ 2 & -3 & 1 \\ \frac{5}{3} & -\frac{5}{2} & \frac{5}{6} \end{vmatrix} = 0$$

sono vettori complanari

c) $\tilde{v} = 2\vec{i} - 3\vec{j} + \vec{k}$ e $\tilde{w} = \vec{i} + h_1\vec{j} + h_2\vec{k}$

$$\tilde{v} \parallel \tilde{w} \iff \tilde{v} \times \tilde{w} = 0 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 1 \\ 1 & h_1 & h_2 \end{vmatrix} = (-3h_2 - h_1)\vec{i}$$

$$-\vec{j}\,(2h_2-1)+\vec{k}\,(2h_1+3)$$

$$\Leftrightarrow \quad \begin{cases} -3h_2-h_1=0 \\ 2h_2-1=0 & h_2=\frac{1}{2} \\ 2h_1+3=0 & h_1=-\frac{3}{2} \end{cases}$$

$$-3\cdot\frac{1}{2}+\frac{3}{2}=0 \quad \text{verificato}$$

Allora $h_1=-\frac{3}{2}$, $h_2=\frac{1}{2}$; $\vec{v}=\vec{i}-\frac{3}{2}\vec{j}+\frac{1}{2}\vec{k}$

2)

$$U=\{(x,y,z),\ x,y,z\in\mathbb{R},\ z^2=0\}$$

$$V=\{(x,y,z),\ x,y,z\in\mathbb{R},\ y=0\}$$

a) U è sottospazio perchè

$U=\{(x,y,0)\in\mathbb{R}^3\}$

Per il II criterio, presi $v_1, v_2\in U$ e $c\in\mathbb{R}$

$v_1=(x_1,y_1,0)$, $v_2=(x_2,y_2,0)$, si ha

$$cv_1-v_2=c(x_1,y_1,0)-(x_2,y_2,0)=$$
$$=(cx_1-x_2,\ cy_1-y_2,\ c\cdot 0-0)=$$
$$=(cx_1-x_2,\ cy_1-y_2,\ 0)\in U$$ perchè la terza componente è nulla

$V=\{(x,0,z)\in\mathbb{R}^3\}$

Per il II criterio, presi u_1 e $u_2\in V$

$u_1=(x_1,0,z_1)$ e $u_2=(x_2,0,z_2)$, $c\in\mathbb{R}$

$$cu_1-u_2=(cx_1-x_2,\ \overbrace{c0-0}^{=0},\ cz_1-z_2)\in V$$

perchè la seconda componente è nulla

b) base di $U=\left[\begin{pmatrix}1\\0\\0\end{pmatrix}\begin{pmatrix}0\\1\\0\end{pmatrix}\right]$ generatori lin. indip. $\begin{pmatrix}\boxed{\begin{matrix}1&0\\0&1\end{matrix}}\\0\quad 0\end{pmatrix}$

$\det\begin{pmatrix}1&0\\0&1\end{pmatrix}=1$

base di $V=\left[\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}0\\0\\1\end{pmatrix}\right]$ generatori lin. indip. $\begin{pmatrix}1&0\\0&0\\0&1\end{pmatrix}$

dim $U = 2$

dim $V = 2$

c) $U + V = \left[\begin{pmatrix}1\\0\\0\end{pmatrix} \begin{pmatrix}0\\1\\0\end{pmatrix} \begin{pmatrix}0\\0\\1\end{pmatrix}\right] = R^3$ è una base perchè $\begin{vmatrix}1 & 0 & 0\\0 & 1 & 0\\0 & 0 & 1\end{vmatrix} = 1$

dim$(U+V) = 3$

$U+V$ coincide con R^3 (sottospazio con la stessa dimensione di R^3)

Siccome

$$\dim \underset{3}{(U+V)} = \dim \underset{2}{U} + \dim \underset{2}{V} - \dim U \cap V$$

$\Rightarrow \dim U \cap V = 1$ Non è somma diretta

3)

$$A = \begin{pmatrix} 1 & -1 & 1 \\ k & 0 & 0 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & 0 \\ -2 & -1 \\ 0 & -k \end{pmatrix}$$

$$C = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \qquad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 1 \end{pmatrix}$$

$$E = AB = \begin{pmatrix} 4 & 1-k \\ 2k & 0 \end{pmatrix}$$

$$F = BD$$

$$= \begin{pmatrix} 2 & 0 & 0 \\ -2 & 3 & -1 \\ 0 & 3k & -k \end{pmatrix}$$

$$G = B^T + C = \begin{pmatrix} 2 & -2 & 0 \\ 0 & -1 & -k \end{pmatrix} + \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

impossibile

$$\det E = -(1-k)\,2k = 2k^2 - 2k = 2k(k-1)$$

E è invertibile per $k \neq 0$ e $k \neq 1$

$$E^{-1} = \frac{1}{2k(k-1)} \begin{pmatrix} 0 & k-1 \\ -2k & 4 \end{pmatrix}$$

4

$$\begin{cases} x - y + 3z = 2 \\ 2x - 3y + z = 5 \\ kx \qquad + 8z = k \end{cases}$$

$$A = \begin{pmatrix} 1 & -1 & 3 \\ 2 & -3 & 1 \\ k & 0 & 8 \end{pmatrix}$$

rango$(A) \geq 2$ $\quad \begin{vmatrix} 1 & -1 \\ 2 & -3 \end{vmatrix} = -3 + 2 = -1$

$$\begin{vmatrix} 1 & -1 & 3 \\ 2 & -3 & 1 \\ k & 0 & 8 \end{vmatrix} = k(-1+9) + 8(-1) = -k + 9k - 8 = 8k - 8$$

$$\begin{cases} \text{rango } A = 3 \text{ per } k \neq 1 \quad (a) \\ \text{rango } A = 2 \text{ per } k = 1 \quad (b) \end{cases}$$

(a) rango$(A\ b)$ = rango A

$\exists!!$ soluzione

$$x = \frac{1}{8k-8} \begin{vmatrix} 2 & -1 & 3 \\ 5 & -3 & 1 \\ k & 0 & 8 \end{vmatrix} = \frac{k(-1+9) + 8(-6+5)}{8(k-1)} = \frac{8k-8}{8k-8} = \underline{1}$$

$$y = \frac{1}{8(k-1)} \begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 1 \\ k & k & 8 \end{vmatrix} = \frac{\overset{40 - k - 32 + 2k - 9k}{(40-k) - 2(16-k) + 3(2k-5k)}}{8(k-1)} = \frac{-8k+8}{8k-8} = -1$$

$$z = \frac{1}{8(k-1)} \begin{vmatrix} 1 & -1 & 2 \\ 2 & -3 & 5 \\ k & 0 & k \end{vmatrix} = \frac{\overset{k \qquad -k}{k(-5+6) + k(-3+2)}}{8(k-1)} = 0$$

(b) per $k=1$

rango $(A \ b) = 2$

$$(A \ b) = \begin{pmatrix} 1 & -1 & 3 & 2 \\ 2 & -3 & 1 & 5 \\ k & 0 & 8 & k \end{pmatrix}$$

Infatti

$$\begin{vmatrix} 1 & -1 & 2 \\ 2 & -3 & 5 \\ 1 & 0 & 1 \end{vmatrix} = (-5+6) + (-3+2) = 1 - 1 = 0$$

Allora si trovano $\infty^{3-2} = \infty^1$ soluzioni

$$\begin{cases} x - y = 2 - 3z \\ 2x - 3y = 5 - z \end{cases}$$

$$x = \frac{\begin{vmatrix} 2-3z & -1 \\ 5-z & -3 \end{vmatrix}}{-1} = \frac{\overset{-6+9z+5-z}{-3(2-3z) + 5 - z}}{-1} = -8z + 1$$

$$y = \frac{\begin{vmatrix} 1 & 2-3z \\ 2 & 5-z \end{vmatrix}}{-1} = \frac{\overset{5-z-4+6z}{5 - z - (4 - 6z)}}{-1} = \frac{5z+1}{-1} = -5z - 1$$

$$\text{Soluzioni} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix} + z \begin{pmatrix} -8 \\ -5 \\ 1 \end{pmatrix}$$

5

$f: R^4 \longrightarrow R^4$ $\quad f(e_1)=e_1,\ f(e_2)=0,\ f(e_3)=e_2$
$f(e_4)=e_3$

$$A=\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

rango $A=3$ $\quad \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}=1$ $\quad$ dim Imm $f=3$

base di Imm $f=\left\{\begin{pmatrix}1\\0\\0\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\\0\end{pmatrix},\begin{pmatrix}0\\0\\1\\0\end{pmatrix}\right\}$

dim Ker f = 4 − dim Imm f = 4 − 3 = 1

$$\text{Ker} f=\left\{\begin{pmatrix}x\\y\\z\\t\end{pmatrix}: \begin{matrix} x=0 \\ z=0 \\ t=0 \end{matrix}\right\}=\left[\begin{pmatrix}0\\1\\0\\0\end{pmatrix}\right]$$ base di Ker f

f non è né iniettiva, né suriettiva (Imm $f \subsetneq R^4$)
(Ker $f \neq \{0\}$)

$f^2: R^4 \longrightarrow R^4$

matrice associata

$$B = A \cdot A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\dim \operatorname{Im} f^2 = 2 \qquad 1 = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \neq 0$

$\dim \operatorname{Ker} f^2 = 4 - 2 = 2$

$\operatorname{Ker} f^2 = \left\{ \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} : \begin{matrix} x = 0 \\ t = 0 \end{matrix} \right\} = \left[\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right]$

6

$$f: \mathbb{R}^3 \longrightarrow \mathbb{R}^3 \qquad B = \{(1,1,0), (1,0,1), (0,0,1)\}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \longrightarrow \begin{pmatrix} x+y+2z \\ 2x+y+3z \\ 3x+y+4z \end{pmatrix} \qquad B' = \{(1,0,-1), (1,0,1), (0,1,0)\}$$

$$M_C^C(f) = \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{pmatrix}$$

$$M_{B'}^{B}(f) = M_{B'}^{C}(i_{\mathbb{R}^3})\, M_C^C(f)\, M_C^{B}(i_{\mathbb{R}^3})$$

$$M_C^{B}(i_{\mathbb{R}^3}) = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$M_{B'}^{C}(i_{\mathbb{R}^3}) = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix}_C = \alpha \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \underset{\downarrow \atop y}{\gamma} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \iff \begin{cases} x = \alpha + \beta \\ y = \gamma \\ z = -\alpha + \beta \end{cases}$$

$$\alpha = x - \beta$$

$$\underline{z = -x + 2\beta}$$

$$\beta = \frac{z+x}{2}$$

$$\begin{cases} \alpha = x - \dfrac{z+x}{2} = \dfrac{x-z}{2} \\ \beta = \dfrac{z+x}{2} \\ \gamma = y \end{cases}$$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}_C = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix}_{B'} \qquad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}_C = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_{B'} \qquad \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}_C = \begin{pmatrix} -\frac{1}{2} \\ \frac{1}{2} \\ 0 \end{pmatrix}_{B'}$$

$$M_{B'}^{B}(f) = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & 1 & 4 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 & 2 \\ 3 & 5 & 3 \\ 4 & 7 & 4 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & -\cancel{\frac{4}{2}}\,2 & -1 \\ 3 & 5 & 3 \\ 3 & 5 & 3 \end{pmatrix}$$

7)

$f: \mathbb{R}^3 \to \mathbb{R}^3$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \to \begin{pmatrix} 2x-2y \\ -3x+3y \\ -z \end{pmatrix} \qquad A = \begin{pmatrix} 2 & -2 & 0 \\ -3 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

$$|\lambda I - A| = \begin{vmatrix} \lambda-2 & 2 & 0 \\ 3 & \lambda-3 & 0 \\ 0 & 0 & \lambda+1 \end{vmatrix} =$$

$$= (\lambda+1)\left((\lambda-2)(\lambda-3)-6\right) =$$

$$= (\lambda+1)\left(\lambda^2-5\lambda+\cancel{6}\cancel{-6}\right) =$$

$$= (\lambda+1)\,\lambda\,(\lambda-5)$$

$\lambda_1 = -1$ m. a. 1

$\lambda_2 = 0$ m. a. 1

$\lambda_3 = 5$ m. a $= 1$

endomorfismo diagonalizzabile per il I criterio

$$V_{-1} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \begin{matrix} -3x+2y=0 \\ 3x-4y=0 \end{matrix} \right\} = \left[\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}\right] \quad 1 = \text{m. g di } \lambda_1$$

$$V_0 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \begin{matrix} -2x+2y=0 \\ 3x-3y=0 \\ z=0 \end{matrix} \right\} = \left[\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}\right] \quad 1 = \text{m. g di } \lambda_2$$

$$V_5 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \begin{matrix} 3x+2y=0 \\ \cancel{3x+2y=0} \\ 4z=0 \end{matrix} \right\} = \left[\begin{pmatrix} 2 \\ -3 \\ 0 \end{pmatrix}\right] \quad 1 = \text{m. g di } \lambda_3$$

$$\mathcal{B} = \left\{ \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 2 \\ -3 \\ 0 \end{pmatrix} \right\} \qquad N = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 1 & -3 \\ 1 & 0 & 0 \end{bmatrix}$$

$$AN = ND \qquad D = \begin{pmatrix} -1 & & \\ & 0 & \\ & & 5 \end{pmatrix}$$

8

$$A = \begin{pmatrix} 10 & 0 & -6 \\ 0 & 4 & 0 \\ -6 & 0 & 10 \end{pmatrix} \qquad q(x,y,z) = 10x^2 + 4y^2 + 10z^2 - 12\,xz$$

$$|\lambda I - A| = \begin{vmatrix} \lambda-10 & 0 & 6 \\ 0 & \lambda-4 & 0 \\ 6 & 0 & \lambda-10 \end{vmatrix} =$$

$$= (\lambda-4)\left((\lambda-10)^2 - 36\right) =$$

$$= (\lambda-4)\left(\lambda^2 + \overbrace{100-36}^{64} - 20\lambda\right)$$

$$= (\lambda-4)^2(\lambda-16)$$

$$\lambda^2 - 20\lambda + 64 = 0$$

$$\lambda = \frac{20 \pm \sqrt{400-256}}{2} = \frac{20 \pm 12}{2} = \begin{cases} 16 \\ 4 \end{cases}$$

$$V_{16} = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \begin{array}{l} 6x + 6z = 0 \\ 6y = 0 \\ \cancel{6x + 6z = 0} \end{array} \right\} = \left[\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right]$$

$$V_4 = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} \begin{array}{l} -6x + 6z = 0 \\ \cancel{6x - 6z = 0} \end{array} \right\} = \left[\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right]$$

base ortonormale $B' = \left\{ \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix} \right\}$

9)

$$V = \left[\begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} \right] \qquad v_1' = v_1 = \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix}$$

$$v_2' = v_2 - \frac{\langle v_2, v_1' \rangle}{\langle v_1', v_1' \rangle} v_1' = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} - \frac{\left\langle \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} \right\rangle}{\left\langle \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} \right\rangle} \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} =$$

$$= \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} - \frac{8+12}{4+36} \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 0 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 2 \\ 6 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}$$

$$v_1'' = \frac{(2,6,0)}{\sqrt{40}} = \left(\frac{2}{\sqrt{40}}, \frac{6}{\sqrt{40}}, 0 \right) = \left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}, 0 \right)$$

$$v_2'' = \frac{(3,-1,0)}{\sqrt{9+1}} = \left(\frac{3}{\sqrt{10}}, -\frac{1}{\sqrt{10}}, 0 \right) \qquad \Rightarrow \mathcal{B}' = \{ v_1'', v_2'' \}$$

Per determinare una base ortonormale di $\mathbb{R}^3$ basta aggiungere $(0,0,1) = v_3''$.